MIL-P-23312C
18 November 1976
SUPERSEDING
MIL-P-0023312B(OS)
23 April 1975

MILITARY SPECIFICATION

PALLETS, MATERIAL HANDLING, METAL (FOR ORDNANCE ITEMS); MARK 3 MOD 0, MARK 12 MOD 0 AND MARK 12 MOD 1

This specification is approved for use by all Department and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers three types of material handling pallets which can be handled by fork trucks, pallet trucks, and bar-type pallet slings. The pallets consist of a steel wire grid deck attached to a sheet steel supporting structure which provides 4-way entry for material handling equipment. (See 6.1).

1.2 Classification. The pallets shall be of the following types, as specified (See 6.2):

Mk 3 Mod 0, 40 by 48 inches
Mk 12 Mod 0, 35 by 45-1/2 inches
Mk 12 Mod 1, 35 by 45-1/2 inches

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-S-781	Steel, Strapping, Flat

STANDARDS

Military

MIL-STD-105	Sampling Procedure and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Division (611), Indian Head, MD 20640 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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DRAWINGS

Naval Sea Systems Command (CODE IDENT. 10001)

DL 564200	Pallet Mk 3 Mod 0
564200	Pallet Mk 3 Mod 0
2086479	Pallet, Material Handling, Mk 12 Mod 0
DL 2086479	Pallet, Material Handling, Mk 12 Mod 0
2645217	Pallet, Material Handling, Mk 12 Mod 1
DL 2645217	Pallet, Material Handling, Mk 12 Mod 1

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions, should be obtained from the procurement activity or as directed by the contracting officer.)

*2.2 Other Publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on the date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E 380-76	Standard for Metric Practice (IEEE-268-1975 ANSI 2210.1)

(Application for copies of the ASTM standards by serial designation and title should be made to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103)

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (See 6.2) before production is commenced, the first articles of Mk 3 Mod 0, Mk 12 Mod 0 and Mk 12 Mod 1 pallets shall be submitted or made ready for examination and testing as specified in 4.3.1.

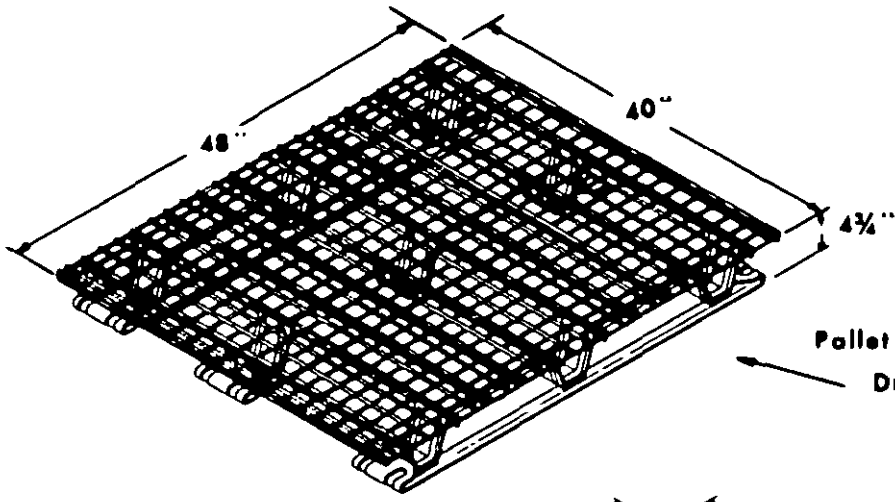
3.2 Materials and components. Materials and components shall conform to all documents listed on the applicable drawings.

3.3 Design and construction. The design and construction of the Mk 3 Mod 0, Mk 12 Mod 0 and Mk 12 Mod 1 pallets shall conform to DL 564200, DL 2086479 and DL 2645217, respectively (see Figure 1).

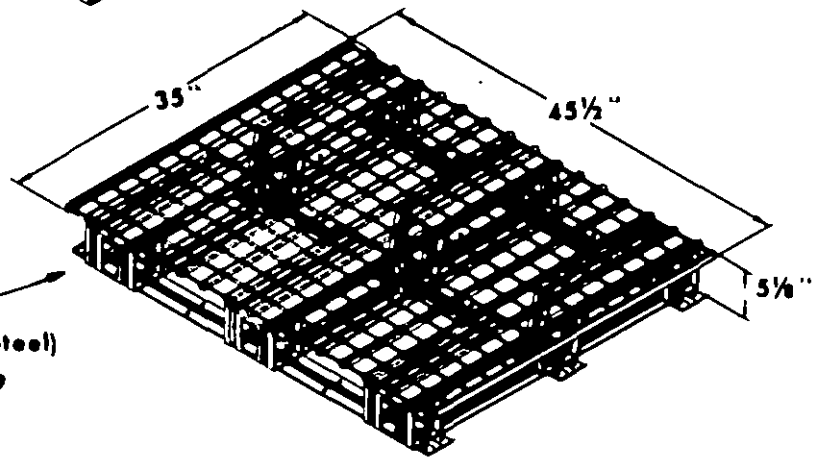
When metric is specified conversion shall be as specified in ASTM E 380-76.

3.3.1 Gages. The supplier shall provide himself with whatever gages are necessary and adequate to ensure that the material to which this specification applies will meet the dimensional requirements shown on the applicable drawings. If so stipulated in the contract, order, or requisition, the Government will furnish drawings or pertinent Navy Final Inspection Gages, when available, for guidance in the design of the supplier's

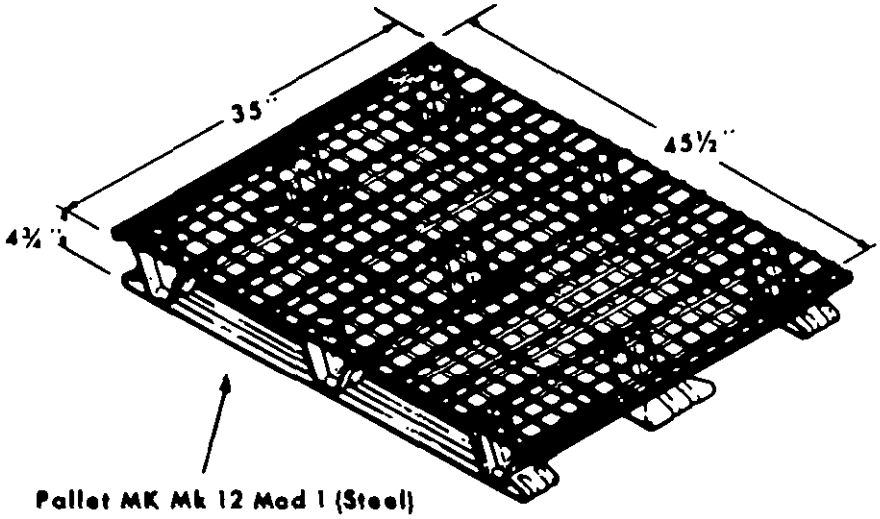
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Pallet Mk 3 Mod 0 (Steel)
Drawing 564200
DL 564200



Pallet Mk 12 Mod 0 (Steel)
Drawing 2086479
DL 2086479



Pallet MK Mk 12 Mod 1 (Steel)
Drawing 2645217
DL 2645217

FIGURE 1. STEEL PALLETS

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inspection gages. However, such procedure shall not relieve the supplier of his responsibility in the design and manufacture of such gages as may be required for the satisfactory fulfillment of the contract requirements, but is intended to facilitate acceptance of the pallet by Navy Final Inspection Gages.

3.3.2 Conflicting requirements. Conflicting requirements arising between this specification and any specification, publication, or drawing listed herein shall be referred in writing to the procuring activity or appointed agent for interpretation, clarification, and resolution (or correction).

3.4 Performance requirements and product characteristics. The pallets shall meet the following performance requirements and product characteristics:

*3.4.1 Weld machine. All machines used for assembly of the pallets shall be qualified for the material and particular combination of thickness prior to being used in production. Qualification shall consist of the production of satisfactory specimens to pass tests in accordance with 4.5.2.

3.4.2 Cross wire weld - shear. When tested as specified in 4.5.2.1, the welds shall not crack nor fail. Failure of any specimen to meet these requirements shall constitute a failure of the qualification. In case of failure the welding method or equipment shall be modified and the test repeated.

3.4.3 Butt weld - tension. When tested as specified in 4.5.2.2, the welds shall not crack nor fail. Failure of any specimen to meet these requirements shall constitute a failure of the qualification. In case of failure the welding method or equipment shall be modified and the test repeated.

3.4.4 Overload test. When tested as specified in 4.5.1.2 the pallets shall be free from any permanent deformation or weld failures.

3.4.5 Push-pull test. When tested as specified in 4.5.1.3, the pallets shall show no weld failure or damage which would affect usability of pallet, and the skids shall not be damaged other than minor scratching or scoring.

3.4.6 Vibration test. When tested as specified in 4.5.1.4 the pallets shall show no permanent deformation or weld failure.

3.4.7 Edgewise-drop (rotational) test. When tested as specified in 4.5.1.5, the pallets shall show no failures or damage which would affect the usability of the pallet.

3.4.8 Cornerwise-drop (rotational) test. When tested as specified in 4.5.1.6, the pallets shall show no weld failures or damage which would affect usability of the pallet.

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3.4.9 Impact test. When tested as specified in 4.5.1.7 the pallets shall show no weld failures or damage which would affect usability of pallet.

3.5 Workmanship. The pallets shall be constructed and finished in a manner to assure compliance with the requirements of this specification. All surfaces and edges shall be free from sharp edges, rough spots, and similar defects that could possibly damage items placed on the pallet or be a hazard to operation personnel when handling.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchse order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The examination and testing of the pallets shall be classified as follows:

(a) First article inspection. First article inspection consists of examinations and tests performed on samples which are representative of the production item after the award of a contract to determine that the production item conforms to the requirements of this specification. (See 3.1, 4.3 and 4.3.1)

(b) Quality conformance inspection. Quality conformance inspection consists of production control tests and examinations performed on individual products or lots to determine conformance of the products or lots with the requirements set forth in this specification (See 4.4 through 4.4.4.3).

4.3 First article inspection. The first article inspection of the pallets shall consist of examinations and tests for all of the requirements of this specification. First article tests shall be accomplished on samples selected as specified in 4.3.1 which are representative of the production of the pallet after the award of the contract to determine that the production meets the requirements of this specification. These tests are detailed in 4.5.1. Acceptance shall be based on no defects in the samples. Failure of the sample to comply with these requirements will result in the rejection of the pallet.

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4.3.1 First article samples. Unless otherwise specified as soon as practicable after the award of the contract or order, the first two pallets of each mark, as applicable, manufactured by the methods and equipment proposed for the production of the pallets shall be submitted for first article tests to determine compliance with the requirements of the contract, specifications and drawings. Further production of the pallet by the supplier prior to approval by the procuring activity or the completion of the tests on the first article samples shall be at the supplier's risk, First article samples accepted will be applied as part of the quantity specified by the contract or order, but shall not be used on any equipment manufactured for the Government. The first article samples shall be forwarded to the address specified in the contract or order (see 6.2.2). The first article samples shall be plainly identified by securely attached tags marked with the following information:

Samples submitted by (name)(date) for first article inspection in accordance with the requirements of MIL-P-23312 under Contract No. (number)

4.3.2 Upon completion of the first article inspection, all the applicable inspection reports and, when applicable, recommendations and comments pertinent for use in monitoring production will be forwarded to the Government quality control representative.

4.4 Quality conformance inspection. The quality conformance inspection shall consist of:

(a) Qualification of weld process or machine (4.4.1)
(b) Production control tests (4.4.3, 4.5.2)
(c) Visual and dimensional examination (4.4.4.1, 4.4.2)
(d) Examination for preparation for delivery (4.4.4.1, 4.4.4.3)

*4.4.1 Qualification of weld process or machine. The weld process or machine shall be qualified under the requirements outlined in 3.4.1, 3.4.2 and 3.4.3.

4.4.2 Sampling for production control tests.

4.4.2.1 Sampling for lot acceptance. The following lot acceptance test shall be performed in accordance with MIL-STD-105.

(a) Visual and dimensional examination (4.4.4.1 and 4.4.4.2) - Inspection Level II Acceptable Quality Level (AQL's) 1% for major defects and 2.5% for minor defects.

(b) Preparation for delivery (4.4.4.1 and 4.4.4.3) Inspection Levels S-2 acceptance number zero.

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4.4.2.2 Production control samples. Sampling for production control shall consist of one cross wire production control coupon and five butt welds taken at the beginning and end of each shift and whenever the electrodes are changed or dressed, or when any changes are made in the control of the machine. Additional samples may be taken by the supplier during the shift to insure against rejection of production pallets.

4.4.2.2.1 Shear test samples. Five shear test specimens shall be selected at random from multiple welding cross wire coupons containing not less than eight consecutive welds. The coupons shall be of the same material and gage combinations which are to be used in the production of the pallets and shall be welded as required in production. The welding machine settings for these coupons shall be recorded.

4.4.2.2.2 Tensile test samples. Two tensile test specimens shall be selected at random from the butt weld samples prepared of the same material and gage combinations which are to be used in the production of the pallets and shall be welded as required in production. The welding machine settings shall be recorded.

4.4.3 Production control tests. Production control tests shall be accomplished on samples selected as specified in 4.4.2.2 which are representative of the production of the pallet to determine that the production meets the requirements of this specification. These tests are detailed in 4.5.2. Acceptance shall be based on no defects in the sample. Failure of the sample to comply with these requirements will result in the rejection of the lot.

4.4.4 Lot acceptance tests. These tests shall be accomplished on pallets of each type being submitted to the Government for acceptance. Inspection and test procedures shall be submitted for Government approval prior to commencement of production.

4.4.4.1 Examinations. The examinations specified in 4.4.4.2 and 4.4.4.3 shall be accomplished on each pallet submitted for acceptance under the contract to ensure that the pallet meets the requirements of 3.2, 3.3 and 3.5. Examinations shall either be performed by the manufacturer and witnessed by the Government Inspector, or they may be conducted by the Government Inspector.

4.4.4.2 Visual and dimensional examination. Each pallet selected in accordance with 4.4.2.1 shall be visually inspected for completeness of manufacture, freedom from defects, proper identification, workmanship, and dimensions.

4.4.4.3 Preparation for delivery. Each fully prepared bundle of pallets selected in accordance with 4.4.2.1 shall be examined to determine conformance to section 5 of this specification.

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4.5 Test methods.

4.5.1 First article tests.

*4.5.1.1 Preparation for testing. All tests shall be performed with the pallet assembled with containers and metal strapping to form a palletized unit load. The containers used shall provide a uniformly distributed load of 4000 pounds covering the entire pallet. The unit load containers may extend beyond the edges of the pallet a distance not exceeding 2 inches.

4.5.1.2 Overload test. The palletized unit load shall be placed on a level surface and an equally distributed load of 16,000 pounds placed on the unit load. After 12 hours, the pallet shall meet the requirements of 3.4.4.

4.5.1.3 Push-pull test. The palletized unit load shall be pushed and towed a distance of 5 feet parallel to the axis of the pallet runners and 5 feet perpendicular to the axis of the pallet runner. After being subjected to this test, the pallet shall meet the requirements of 3.4.5.

4.5.1.4 Vibration test. The palletized unit load shall be placed on a vibration platform. The motion of the platform shall be such that any point on the platform moves in a vertical linear path with a total excursion of 1 inch. The frequency of the motion shall result in the pallet leaving the platform as determined by withdrawal of a paper sheet from under the runners of the pallet, approximately 250 to 270 revolutions per minute. After vibrating for 2 hours, the pallet shall meet the requirements of 3.4.6.

4.5.1.5 Edgewise-drop (rotational) test. The palletized unit load shall be placed with one end of the pallet supported on a sill nominally 6 inches high. The unsupported end of the pallet shall then be raised to a height of 12 inches and allowed to fall freely to the concrete pavement or similarly unyielding surface. This test shall be applied once to each end of the pallet. After repeating this test on all ends the pallet shall meet the requirements of 3.4.7.

4.5.1.6 Cornerwise-drop (rotational) test. The palletized unit load shall be subjected to four drop tests. One corner of the pallet shall be supported on a block nominally 6 inches in height and a block nominally 12 inches in height shall be placed under the other corner of the same end. The unsupported end of the pallet shall be raised so that the lower corner of that end reaches a height of 18 inches and then allowed to fall freely to the concrete pavement or similarly unyielding surface. After repeating this test on all four corners the pallet shall meet the requirements of 3.4.8.

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4.5.1.7 Impact test. The palletized unit load shall be impacted at a velocity of 10 feet per second against a rugged unyielding surface. This test may be performed by pendulum impact, incline impact, or any applicable method. After impacting the unit load on all four normally vertical faces, the pallet shall meet the requirements of 3.4.9.

4.5.2 Production control tests. Production control samples selected as specified in 4.4.2.2 shall be subjected to the following tests.

4.5.2.1 Shear test. The weld specimens shall be loaded to place the welds in shear as specified on Drawings 564200, 2086479 or 2645217, as applicable. Welds tested shall meet the requirements of 3.4.2.

4.5.2.2 Tensile test. The weld specimens shall be loaded to place the welds in tension as specified on Drawings 564200, 2086479 or 2645217, as applicable. Welds tested shall meet the requirements of 3.4.3.

5. PACKAGING

5.1 Packing. Packing shall be level B or C as specified in the contract or purchase order.

5.1.1 Level B. Unless otherwise specified in the contract, requisition or order, the pallets shall be stacked in bundles of eight pallets of the same type and strapped together with four 0.75- x 0.020-inch steel straps conforming to QQ-S-781, Class 1, Type I or IV, Finish B.

5.1.2 Level C. The pallets shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity. The packs shall comply with common carrier regulations applicable to the mode of transportation.

5.2 Marking. In addition to any special marking, required by the contract or order, the bundles or packs shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES AND CONCLUDING MATERIAL

6.1 Intended use. The pallets are to be used for the construction of palletized unit loads for the handling, storage and transportation of ordnance items in intradepot, interdepot and shipboard operations.

*6.2 Ordering data. Procurement data should specify the following:

(a) Title, number and date of this specification
(b) Quantity
(c) Mark and Mod designation of pallets being purchased
(d) Whether first article inspection is waived (See 4.3.1)

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(e) Specify address of activity which will receive first article inspection
(f) Exceptions to this specification
(g) Applicable drawings and other documents
(h) Selection of applicable level of packing (See 5.1)

6.3 Criteria for use of the proper level of packing shall be as follows:

6.3.1 Level B. This level shall be used for pallets which will be subjected to multiple domestic shipments, handling or storage.

6.3.2 Level C. This level shall be used for pallets which are to be shipped to domestic installations for immediate use at the first receiving activity.

*6.4 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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DOCUMENT IDENTIFIER AND TITLE

MIL-P-23312C(OS) "Pallets, Material Handling, Metal (For Ordnance Items);

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

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A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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